

Vector Analysis

Problem 1.1

Problem 1.2

Is the cross product associative?

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \stackrel{?}{=} \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

Solution:

No, for example set

$$\begin{aligned}\mathbf{A} &= \hat{x} \\ \mathbf{B} &= \hat{x} \\ \mathbf{C} &= \hat{z}\end{aligned}$$

Then, the left hand side above becomes

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\hat{x} \times \hat{x}) \times \hat{z} = 0$$

while the right hand side is

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \hat{x} \times (\hat{x} \times \hat{z}) = \hat{x} \times -\hat{y} = -\hat{z}$$

□

Problem 1.3

Solution:

□

Problem 1.4**Problem 1.5**

Prove the **BAC-CAB** rule,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

by writing out both sides in component form.

Solution:

Writing out the vectors in component form

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

$$\mathbf{B} = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}$$

$$\mathbf{C} = C_x \hat{\mathbf{x}} + C_y \hat{\mathbf{y}} + C_z \hat{\mathbf{z}}$$

starting with lhs

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = (B_y C_z - B_z C_y) \hat{\mathbf{x}} + (B_z C_x - B_x C_z) \hat{\mathbf{y}} + (B_x C_y - B_y C_x) \hat{\mathbf{z}}$$

rewriting with a new matrix $\mathbf{D}$ where

$$D_x = B_y C_z - B_z C_y$$

$$D_y = B_z C_x - B_x C_z$$

$$D_z = B_x C_y - B_y C_x$$

gives

$$\mathbf{D} = \mathbf{B} \times \mathbf{C} = D_x \hat{\mathbf{x}} + D_y \hat{\mathbf{y}} + D_z \hat{\mathbf{z}}$$

Now

$$\begin{aligned} \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{A} \times \mathbf{D} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ D_x & D_y & D_z \end{vmatrix} = \\ &= (A_y D_z - A_z D_y) \hat{\mathbf{x}} + (A_z D_x - A_x D_z) \hat{\mathbf{y}} + (A_x D_y - A_y D_x) \hat{\mathbf{z}} \end{aligned}$$

RHS:

$$\mathbf{A} \cdot \mathbf{C} = A_x C_x + A_y C_y + A_z C_z = P$$

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z = Q$$

$$\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) = B_x P \hat{\mathbf{x}} + B_y P \hat{\mathbf{y}} + B_z P \hat{\mathbf{z}}$$

$$\mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = C_x Q \hat{\mathbf{x}} + C_y Q \hat{\mathbf{y}} + C_z Q \hat{\mathbf{z}}$$

$$\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = (B_x P - C_x Q) \hat{\mathbf{x}} + (B_y P - C_y Q) \hat{\mathbf{y}} + (B_z P - C_z Q) \hat{\mathbf{z}}$$

Now final step is checking if

$$1: A_y D_z - A_z D_y = B_x P - C_x Q$$

$$2: A_z D_x - A_x D_z = B_y P - C_y Q$$

$$3: A_x D_y - A_y D_x = B_z P - C_z Q$$

For 1:

$$\begin{aligned} \text{LHS: } & A_y(B_x C_y - B_y C_x) - A_z(B_z C_x - B_x C_z) \\ &= A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z \\ &= B_x(A_y C_y + A_z C_z) - C_x(A_y B_y + A_z B_z) \end{aligned}$$

$$\begin{aligned} \text{RHS: } & B_x(A_x C_x + A_y C_y + A_z C_z) - C_x(A_x B_x + A_y B_y + A_z B_z) \\ &= B_x A_x C_x + B_x(A_y C_y + A_z C_z) - C_x A_x B_x - C_x(A_y B_y + A_z B_z) \\ &= B_x(A_y C_y + A_z C_z) - C_x(A_y B_y + A_z B_z) = \text{LHS } \checkmark \end{aligned}$$

For 2:

$$\begin{aligned} \text{LHS: } & A_z(B_y C_z - B_z C_y) - A_x(B_x C_y - B_y C_x) \\ &= A_z B_y C_z - A_z B_z C_y - A_x B_x C_y + A_x B_y C_x \\ &= B_y(A_z C_z + A_x C_x) - C_y(A_z B_z + A_x B_x) \end{aligned}$$

$$\begin{aligned} \text{RHS: } & B_y(A_x C_x + A_y C_y + A_z C_z) - C_y(A_x B_x + A_y B_y + A_z B_z) \\ &= B_y A_y C_y + B_y(A_x C_x + A_z C_z) - C_y A_y B_y - C_y(A_x B_x + A_z B_z) \\ &= B_y(A_x C_x + A_z C_z) - C_y(A_x B_x + A_z B_z) = \text{LHS } \checkmark \end{aligned}$$

For 3:

$$\begin{aligned} \text{LHS: } & A_x(B_z C_x - B_x C_z) - A_y(B_y C_z - B_z C_y) \\ &= A_x B_z C_x - A_x B_x C_z - A_y B_y C_z + A_y B_z C_y \\ &= B_z(A_x C_x + A_y C_y) - C_z(A_x B_x + A_y B_y) \end{aligned}$$

$$\begin{aligned} \text{RHS: } & B_z(A_x C_x + A_y C_y + A_z C_z) - C_z(A_x B_x + A_y B_y + A_z B_z) \\ &= B_z A_z C_z + B_z(A_x C_x + A_y C_y) - C_z A_z B_z - C_z(A_x B_x + A_y B_y) \\ &= B_z(A_x C_x + A_y C_y) - C_z(A_x B_x + A_y B_y) = \text{LHS } \checkmark \end{aligned}$$

Since all three components are equal, the BAC-CAB rule is proven. $\square$

Problem 1.6

Problem 1.7

Problem 1.8

Problem 1.9

Problem 1.10

Problem 1.11

Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$

(b) $f(x, y, z) = x^2 y^3 z^4$

(c) $f(x, y, z) = e^x \sin(y) \ln(z)$

Problem 1.12

Problem 1.13

Problem 1.14

Problem 1.15

Problem 1.16

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